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The Laws of Motion!

PART=IA

3.1 Effects of a force: When a force is applied on an object it can produce two main outcomes or effects.

Effect 1: It can change the state of motion or rest of the object.

Effect 2: It can change the size or shape of the body.

Note: A force that is applied to a object with a rigid structure does not change the intermolecular space therefore it does not change the shape of rigid structures. Whereas force can re-shape a non-rigid structure as it can change the intermolecular space.

Force: A force is that physical cause which changes (or tends to change) either the shape or size or the state of motion or rest.

Forces are of 2 types: (i) Contact forces and (ii) Non-Contact forces.

P.T.O

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3.2 Contact Forces :

Def: The forces which are applied on bodies by making a physical contact with them are called contact forces.

Examples of Contact Forces:

Qn. 1 → **Frictional Forces**: When two surfaces slide against each other, a force starts acting on the body in a opposite direction to the motion of the body.

Qn. 2 → **Normal reaction force**: Normal reaction force that an object applies on any body resting on it to support it.

Qn. 3 → **Tension force as applied through strings**:

Suppose an object is hanging in air and ~~its~~ the reason it does not fall is because it is attached to a string and the string act as normal reaction force inturn supporting the block.

Qn. 4 → **Force exerted by a spring**: Consider a spring with its one end kept fixed. If you stretch it and then released it then it will again come back to its original place by applying a suspension force.

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→ **Force inertia during a collision**:
Lets take an example of two cars which have collided. Here car A exerts a force of its momentum on car B and at the same time car B exerts a force on car A of its momentum which causes both of them to displace.

Ex. 5

3.3 Non-Contact Forces:

Def: The forces experienced by bodies even without being physically in touch, are called non-contact forces or forces at a distance.

Examples of Non-Contact forces:

→ **Gravitational forces**: In our universe every object has its own gravitational force. The gravitational force pulls others object and it does not have to be in contact with the object that's it's pulling to pull it that's why its a non contact force.

Ex. 1

→ **Electrostatic force**: Any two like or same charges repel each other and two unlike or dissimilar charges ~~attract~~ attract each other. The force of this this attraction is known as electrostatic force.

Ex. 2

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→ **Magnetic Force**: When two unlike poles of a magnet are put close together, they attract each other and this attraction forces known as Magnetic force.

Imp. 3

Characterstics of non-contact forces:

→ The Gravitational force is always attractive in nature whereas, electrostatic and magnetic forces can either attract or repel.

→ The magnitude of non-contact forces between two bodies depends on the distance of seperation between them. It decreases with increase in seperation and increases with decrease in seperation. So seperation ∝ magnitude.

Important

Exercise - 3(A)

1)

(a) Contact forces are those that are applied on bodies when in physical contact. For example for friction to occur, two surfaces must physically come in contact. And an other example is collision force. Here for object A to apply a force on object B they must physically collide.

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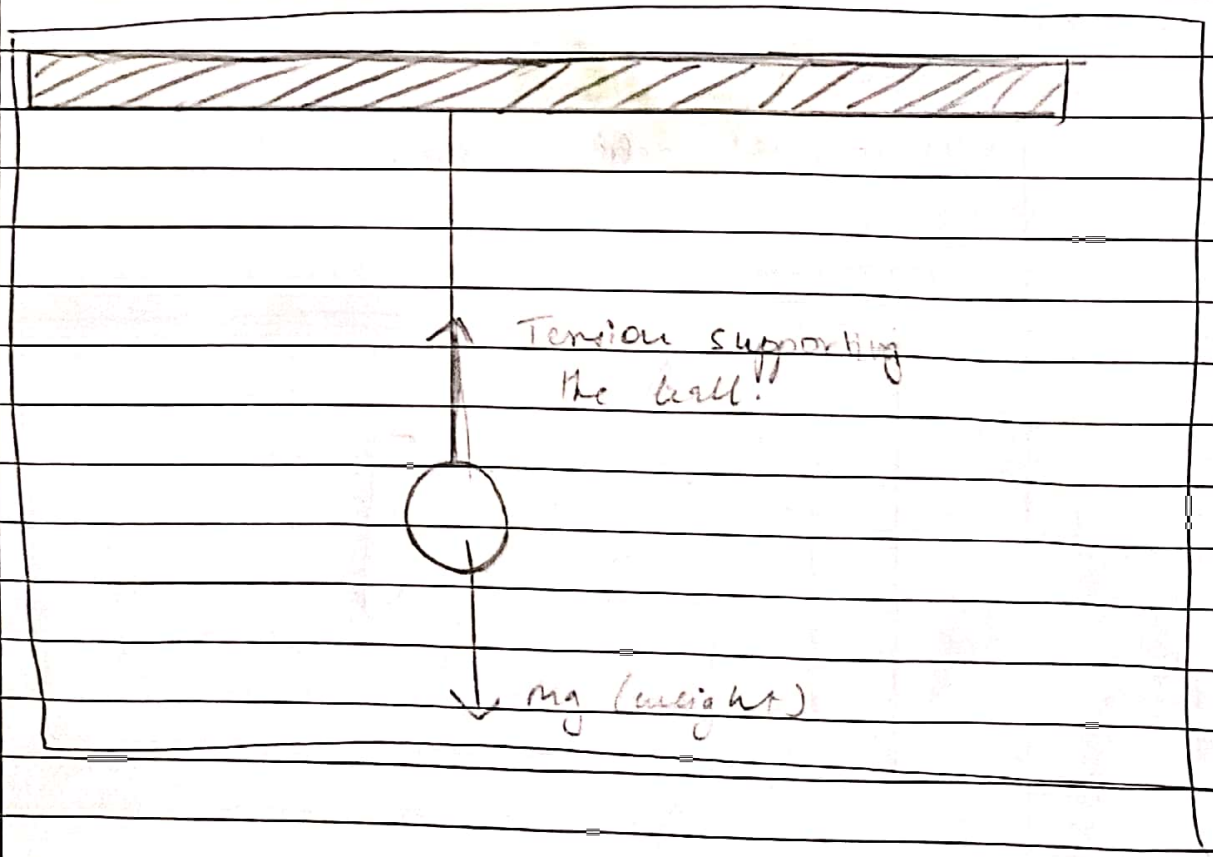
1b) Non-Contact Forces are the forces experienced by object even without being physically in touch. Ex - Gravity pulls without touching a object and two magnets attract each other without physically touching each other.

2)

Contact - Force	Non - Contact Force
(a) Frictional force	(d) gravitational force
(b) normal reaction force	(e) electrostatic force
(c) forced tension on a string	(f) magnetic force.

3) (a) Friction
(b) Magnetic force.

4)
a)



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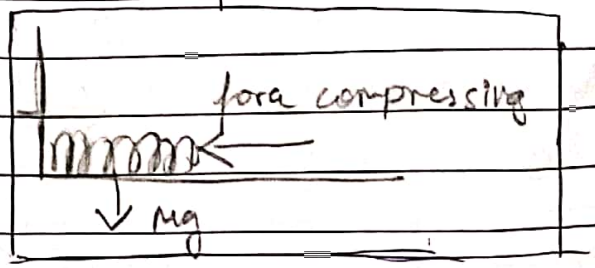
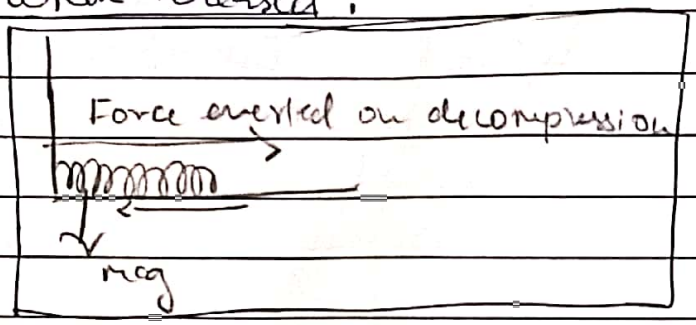
(b) When Compressed:



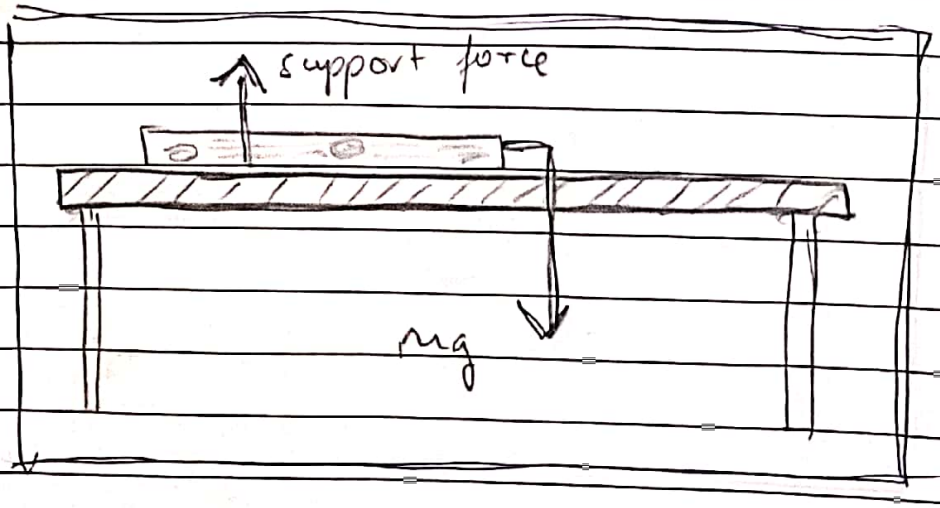
When released:



(c)

(a) gravity
(b) Normal reaction force.

Diagram:



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6) The Forces would increases as Force ∝ distance and it would increases the force by 4 times (2 side x 2).

7) (i) the shape changes as the intermolecular space could change
(ii) The shape of the rigid body does not change as their is very less spaced to be changed.

The two effects differ as the shape in non-rigid structures change whereas the shape in non-rigid structures do not change.

8 a)
(a) de acceleration force or break
(b) acceleration
(c) compression

MCQs

1) (c)
2) (b)

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